

Dachuan Song

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Google Scholar | GitHub | LinkedIn | Personal Website

Education

George Mason University, Fairfax, Virginia, USA Jan 2024 – Present

College of Engineering and Computing

Ph.D. in Electrical and Computer Engineering, Current GPA: 3.91

Advisor: Prof. Xuan Wang

- **Major Courses:** *Linear Systems, Random Processes, Kalman Filtering, Distributed Control and Optimization, System Identification, Advanced learning from data*

University of Southampton, Southampton, United Kingdom

Sept 2022 – Sept 2023

School of Electronics and Computer Science

M.Sc. in Computer Science, GPA: 3.8 (Top 3%)

- **Major Courses:** *Advanced Machine Learning, Advanced Deep Learning, Computer Vision, Bayesian, Active & Reinforcement Learning*

Xinjiang University, Xinjiang, China

Sept 2017 – Jun 2021

School of Software

B.Eng. in Software Engineering, GPA: 3.4 (Top 5%)

Publications

CONFERENCE PAPERS AND PREPRINTS:

Elastic Spectral State Space Models for Train-Once Budgeted Inference 2026

Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang

<https://arxiv.org/abs/2601.22488>

Accepted to *NeurIPS 2026 Main Track (Poster)*; proceedings forthcoming. *arXiv:2601.22488, 2026.*

Elastic Selective Spectral Hybrids for Train-Once, Export-Many Budgeted Inference Sep 2026

Dachuan Song, Chuchu Chen, Xuan Wang

<https://arxiv.org/abs/2609.32486>

Mergeable Model-Side Aggregation States for Long-Context Language Models Jul 2026

Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang

<https://arxiv.org/abs/2607.26448>

Causality-based Subject and Task Fingerprints using fMRI Time-series Data Nov 2024

Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang

<https://doi.org/10.1145/3698587.3701342>

ACM BCB '24, Article 13, pp. 1–10 (Oral Presentation).

JOURNAL ARTICLES:

Reconstructing brain causal dynamics for subject and task fingerprints using fMRI time-series data Oct 2025

Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang

<https://doi.org/10.1007/s13755-025-00388-w>

Health Information Science and Systems 13, Article 70 (2025).

Research Projects

Spectral-Delta-Cache (SDC): Attention-Free Language Models with an Exact-Recall Index

Sep 2026 – Present

- Built a gated exact-recall index that lets attention-free delta-rule models retrieve what followed the latest longest-matching n-gram, using one sort for training and a dictionary for decoding.
- At 155M parameters after 1B tokens, achieved 25.3 validation perplexity (vs. 25.7–25.9 with two attention layers) and 0.99 needle recall at 16K tokens (vs. below 0.05 for baselines).

Elastic Selective Spectral Hybrids for Budgeted Inference

Jul 2026 – Sep 2026

- Gave each spectral channel a recurrent state with input-dependent decay and write/read gates, so it can retain relevant context and forget distractions. Sliding-window attention handles recent tokens.
- Trained channel count and feed-forward width as capacity tiers, producing compact exports from one checkpoint. At 1.53B parameters, fused batch-one decoding took 1.37 ms/token on a B300, 2.14–2.80× faster than the tested Mamba-2 and Mamba-3 implementations at matched parameter counts.

Elastic Spectral State Space Models for Train-Once Budgeted Inference

Jan 2026 – May 2026

- One training run yields independently deployable models for different budgets. Direct truncation removes ordered Hankel channels and their parameters from each smaller export.
- Input-adaptive gates tune channel contributions; budget dropout trains every retained prefix. Across language, audio, long-range, and offline reinforcement-learning tasks, quality changes smoothly as the budget shrinks.

Mergeable Model-Side Aggregation States for Long-Context Language Models

Jul 2026 – Aug 2026

- Added a 2 KiB mergeable HyperLogLog state beside a frozen language model to track distinct counts and set evidence across context chunks.
- Reached 99.2% accuracy on 3,969 aggregate-then-reason tasks with Gemma 4 (31B), without rereading the full history.

Machine Learning for Brain Activity Patterns using EEG (Master's Project)

Apr 2023 – Oct 2023

- Combined CNN spatial features with LSTM temporal modeling for EEG brain-pattern recognition.
- Used autoencoder pretraining, attention, and transfer learning to extract features from limited EEG data.

Internship Experience

Machine Learning Engineer Intern, Optosurgical

May 2026 – Aug 2026

- Recognized surgical phases and instrument actions in video, providing procedure-state signals to downstream systems.
- Compressed and fine-tuned Depth Anything 3 to 29% of its original size while retaining near-full depth-estimation performance.

Technical Skills

- **Architectures & Inference:** spectral and selective state space models, recurrent memory, sliding-window attention, long-context aggregation, multi-budget model export.
- **ML Engineering:** PyTorch, CUDA, FFT convolution, multi-GPU training, model compression, fused recurrent decoding, reproducible evaluation.
- **Modeling & Control:** linear systems, Kalman filtering, system identification, LQR/MPC, autoregressive models.